

Quiz Master App Documentation

Project Overview

This Quiz Master application is developed using Kotlin, Jetpack Compose, Room Database, and MVVM Architecture. The application allows users to participate in quizzes from different categories, answer multiple-choice questions, review answers, and track their quiz performance through score history.

Project Scope

The application provides an interactive quiz platform where users can select categories, answer questions, view results, and review their performance. Questions and scores are stored locally using Room Database.

Technologies Used

- Kotlin
- Jetpack Compose
- Room Database
- MVVM Architecture
- StateFlow / Compose State
- Navigation Compose

Features

- Category-Based Quizzes
- Multiple Choice Questions
- Score Calculation
- Review Answers
- Score History Tracking
- Room Database Integration
- Dynamic UI Updates

Core Functionalities

1. Quiz Attempt: Users can select a category and answer quiz questions.
2. Read: Display questions stored in the Room Database.
3. Review Answers: Show correct answers and compare them with user responses.
4. Score Calculation: Calculate the user's final score and percentage.
5. Score History: Store and display previous quiz attempts.
6. Room Database Integration: Manage questions and scores locally.

State Management

The UI automatically updates whenever quiz progress, selected answers, questions, or scores change. State management ensures synchronization between the ViewModel, Room Database, and the user interface.

Application Workflow

1. User opens the application.
2. User selects a quiz category.
3. Questions are loaded from Room Database.
4. User answers the questions.
5. Application evaluates the answers.
6. Final score and percentage are calculated.
7. Results are displayed on the Result Screen.
8. User can review answers or view score history.

Screens

Home Screen

- Start Quiz
- View Score History
- Navigate to Categories

Category Screen

- Select Quiz Category
- Start Quiz

Quiz Screen

- Display Questions
- Select Answers
- Navigate Through Questions
- Submit Quiz

Result Screen

- Display Final Score
- Display Percentage
- Review Answers

Review Screen

- Display Correct Answers
- Compare User Answers

Score History Screen

- Display Previous Scores
- Clear History

Architecture

UI Layer (Jetpack Compose)

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ViewModel Layer

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Repository Layer

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Room DAO Layer

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Room Database

Advantages

- Interactive learning experience
- Offline functionality using Room Database
- Fast local data access
- Clean MVVM Architecture
- Easy to maintain and extend

Future Enhancements

- Timer-Based Quizzes
- Difficulty Levels
- Leaderboard System
- User Authentication
- Online Question Sync using Firebase
- Dark Mode Support

Conclusion

The Quiz Master App provides an engaging platform for learning and self-assessment. It demonstrates Room Database integration, MVVM architecture, state management, navigation, and modern Android development practices using Jetpack Compose and Kotlin.